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VoxMentor: A Voice-Driven AI Platform for Interview Simulation, Company-Specific Preparation, and Student Learning

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Abstract - Preparing for interviews requires environments that mirror real-world conditions, including spoken interaction, adaptive questioning, and actionable feedback. Yet, most existing mock interview platforms remain restricted to static question banks or text-based interfaces, offering limited support for communication and holistic skill assessment. This paper introduces VoxMentor, a voice-driven AI platform for immersive interview simulation, coding assessments, and group discussion (GD) practice. The system integrates automatic speech recognition (Whisper), large language models (Google Gemini), and automated coding evaluation (Judge0) into a modular workflow comprising authentication, requirement gathering, virtual interview room setup, real-time Q&A simulation, automated feedback generation, and dashboard visualization. Auxiliary modules further support aptitude testing and company-specific preparation through curated repositories of prior interview questions. In a user study with 30 students, VoxMentor achieved a 6.5%-word error rate, maintained 1.8-second response latency, and improved student satisfaction by 18% compared to text-based baselines. These findings underscore VoxMentor's potential to deliver realistic, scalable, and personalized training for both general readiness and targeted recruitment preparation.

Index Terms— Mock Interviews, Group Discussion, Voice Interaction, Speech Recognition, Intelligent educational systems, e-learning, Large Language Models, AI Feedback Systems, Company-Specific Preparation.

I. INTRODUCTION

Transformation between student life and professional work is one of the most difficult periods in the lives of students. The interviews during the recruitment process are a critical constituent in this transition because it is the main tool employers use to assess the soft skills and technical ability of the candidates. Although the use of the conventional forms of resources (i.e., career counseling, mock interviewing, digital tutorials, and others) can help improve the preparedness of students in high-stakes interviews, not all students are ready to compete in such vocation-defining encounters. They may be very common issues such as a lack of confidence, impaired communication abilities, a deficiency in the technical practices, and exposure to the company-specific set of expectations [1],[2].

Advancements in artificial intelligence (AI), natural language processing (NLP), speech recognition, and conversational agents have enabled the creation of intelligent systems that provide personalized, interactive learning experiences [4],[6]. When it comes to career readiness, the same technologies allow simulating real-life interviews, capturing and processing candidate responses, and providing structured feedback [7],[8]. The voice-driven AI-based platforms can also provide individuals with dynamic, real-time simulations of those real-world situations which can create a necessary sense of pressure and unpredictability that a voice-based AI platform might not be able to simulate adequately.

The paper fills these gaps by introducing VoxMentor, a voice driven artificial intelligence-based interview practice tool and company specific interview preparation tool, as well as a student learning program. The system leverages speech-to-text processing, machine learning–based evaluation models, and large language models (LLMs) to assess verbal fluency, clarity, and content relevance in student responses [11],[12]. The preparation module of VoxMentor is also unique in that it includes company-specific/industry partner frequently asked questions in the simulation preparation process, further aligning training and industry-based expectations [13].

Besides soft-skill measurement, VoxMentor also has technical practice section, including coding tests, aptitude exams, supported through online code editors and execution [12]. This multimodal variance makes students holistically trained to improve on their proficiency in communication and their technical preparedness [14],[15].

The primary contributions of this work are as follows:

I. Creation of a voice feedback Tutor that explains answers in real time during the interview process using AI and NLP to provide institutional feedback instantly.

II. Implementation of a company tailored preparation framework, that curates and adjust recruiter driven question sets related to specific learning.

III. A fully integrated skill evaluation climate, both technical problem-solving and communication skills assessment, is integrated.

IV. Implementation of a group discussion (GD) simulation module that evaluates participation balance, content relevance, and teamwork in multi-user sessions.

By closing the gap between academic training and industry needs, VoxMentor reveals in a powerful way how AI-driven platforms can transform student career readiness, offering a sustainable, responsive, and smart formula to interviews training within higher education facilities [3],[5],[15].

II. LITERATURE SURVEY

Preparing to an interview has taken a new turn, and traditional mock interviews have been replaced by intelligent AI-based systems with the ability to mimic the actual situation and give meaningful feedback. The old career counseling practice and question banks cannot keep pace with the dynamics of the new and highly structured hiring efforts. Recent developments leverage Artificial Intelligence (AI), Natural Language Processing (NLP), and speech recognition to create adaptive, voice-driven learning environments for students.

Li et al. [1] introduced Ezinterviewer, an AI-based mock interview generator, demonstrating the utility of adaptive question generation for enhancing interview readiness. Extending this concept, Pawar et al. [2] proposed a comprehensive AI-driven approach for interview preparation, integrating simulation with structured feedback, while Sivaramakrishnan et al. [3] developed a CNN-based real-time evaluation method for candidate performance. Daryanto et al. [4] emphasized the value of reflective learning through dialogic feedback in simulated interview environments, and Amrutha et al. [5] incorporated emotion and confidence analysis to provide multidimensional evaluation. In parallel, Koshti et al. [6] presented an integrated system combining resume analysis, HR simulation, and technical assessments, whereas Gorer and Aydemir [7] investigated robotic and virtual tutors as emerging modalities for interview training. Together, these demonstrations present the possibility that AI can make interview preparation a data-driven, adaptive, and scalable learning experience that is not subjectively based on the resources available to the learner.

A second category of work has been on multimodal assessment and transcription tools. Early contributions by Naim et al. [8] established the predictive role of linguistic and paralinguistic features in determining interview performance. More recent efforts have expanded these capabilities with deep learning and affective computing [3], [5]. Since voice interactions are central to interviews, the reliability of automatic speech recognition (ASR) remains a foundational element. Eftekhari [9] explored intelligent transcription systems in qualitative research, demonstrating practical applications of ASR, while Radford et al. [10] introduced Whisper, a large-scale, weakly supervised ASR model capable of robust multilingual and accent-independent transcription. The above technologies make real-time, accurate transcriptions in VoxMentor possible.

The other developing trend entails the preparation of interviews in relation to the particular organizational setting. Sharma et al. [11] proposed retrieval-augmented generation (RAG) for domain-specific question answering, demonstrating how knowledge retrieval can condition question generation on company-related data. The application of such methods is most appropriate in facilitation of firm-specific preparation in the case of VoxMentor, where candidates can train themselves in questions specific to Google or Amazon.

Pertinent is the incorporation of technical assessment. Tong et al. [12] introduced CODEJUDGE, a benchmarking framework that uses large language model (LLM)-based judges to evaluate coding tasks, providing a scalable solution for aptitude and programming assessment. This will make AI-based interviewing solutions even more comprehensive since it is not limited to behavioral and communication components only.

Lastly, it has been demonstrated that visual feedback of results and progress as provided by the learning analytics dashboards can meaningfully enhance student engagement as well as self-regulated learning. Susnjak [13] demonstrated that dashboards provide actionable insights that guide learners toward targeted improvements, while Duan et al. [14] confirmed their positive impact on self-regulated learning outcomes in higher education contexts. Complementary to this, Chopra and Haaland [15] analyzed the methodological and ethical considerations of conducting AI-driven interviews, emphasizing the importance of fairness and transparency in virtual interviewing systems.

III. ARCHITECTURE

In order to present an adaptive, voice-based interview simulation program, the proposed system architecture of VoxMentor has four functional modules. In it, every module selectively acts in a narrow, predetermined operation and passes the result onto the next stage. The system details are calculated in such a way that it gives us a complete chain of speech input to customized feedback.

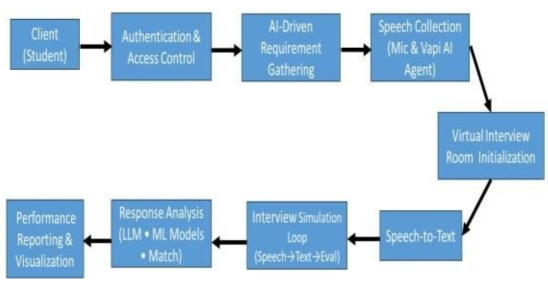

Fig 1. Proposed System

The four main types of modules are:

Type 1: Speech Collection

Type 2: Speech-to-Text Processing

Type 3: Response Analysis and Company-Specific Matching

Type 4: Feedback Generation and Visualization

3.1 Speech Collection

Speech Collection module will be the container of entry way of the system. The platform is interacted through the client devices by students and these are web applications or mobile interfaces. Audio is captured in real-time through microphone input and subjected to preprocessing steps, including noise reduction, echo cancellation, and voice activity detection (VAD). This will optimize the signal captured so that it can facilitate successful speech recognition tasks when the acoustic conditions are unfavorable. The collected audio streams are securely transmitted to the server using encrypted channels (e.g., WebRTC over TLS) to maintain both confidentiality and low latency in communication.

3.2 Speech-to-Text Processing

The Automatic Speech Recognition (ASR) module converts raw audio streams into textual transcripts. VoxMentor supports integration with OpenAI Whisper, the most up-to-date ASR model that has been trained on cross-lingual and cross-task speech data. The system is resistant to speaker variability, ambient noise and accents, which enables it to handle a wide range of student diversity. This layer also performs timestamped transcription, enabling the synchronization of speech with prosodic features (e.g., pauses, intonation). These transcripts serve as the primary input for the Natural Language Processing (NLP) pipeline, thus forming the backbone of semantic analysis.

3.3 Company-Specific Matching

VoxMentor includes a stapled interview questions database where the information is collected by drawing upon widely available company interviews, HR principles, and technical industry-related questions. The system provides adaptive follow-up questions so that each organization is simulated as recruiter by adding specific contextual information. A similarity matching algorithm (e.g., cosine similarity over embeddings) evaluates how closely a student's response aligns with model answers expected in that company context.

3.4 Feedback Generation and Visualization

The final stage of the workflow is the generation of multi-dimensional feedback that integrates results from all preceding modules to provide a holistic view of learner performance. Feedback is divided into two categories. Soft-skill measures evaluate oral fluency, articulation, pacing, and the use of filler words, with confidence and fluency trends presented through visualizations such as radar plots and progression charts. Technical-skill metrics are derived from coding responses executed in sandboxed environments using the Judge0 API, which assesses correctness, execution efficiency, and time complexity.

All results are consolidated in a student dashboard that records longitudinal performance across multiple sessions. This enables both learners and instructors to monitor progress, identify strengths, and track areas for improvement over time. In addition, structured feedback reports can be exported for use in academic evaluation, career counseling, or institutional readiness assessment.

IV. SYSTEM IMPLEMENTATION

The given system, VoxMentor, can be described as a modular tool based on the speech-driven AI, cloud technology, and intelligent assessment programs and imitates the real interview process. Its methodology follows a repetitive two-category structure where the core simulation loop, which combines an end-to-end recording and refinement of the actual mock interview procedure itself, and autonomous auxiliary modules, that provide additional practice in aptitude/coding as well as external company related knowledge. This division clarifies the goals of the system, though as the systems grow in the future this will change.

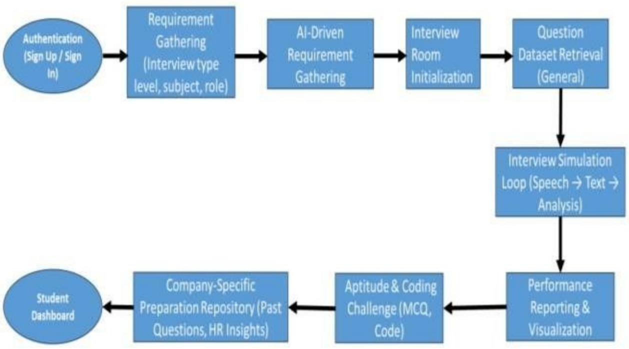

Fig.2 proposed Methodology

4.1 User Authentication and Access Control

It will start with secure user authentication that will be achieved through Firebase Authentication, it includes sign-up and sign-in processes. All of the credentials are encrypted and authenticated via backend API routes, making compliance with current authentication standards. Upon success the candidate is redirected on to the dashboard, with identification of a secure session to have individual interactions.

4.2 AI-Driven Requirement Gathering

Following the login, an AI-based conversation module captures candidate-specific needs, including the type of interview (technical, HR or behavioral), subject focus (e.g., SQL, Java, Python) and degree of difficulty (basic, intermediate, or advanced). This will make the interviewing process more personalized to the goals of the candidate contributing to a greater realistic and relevant process.

4.3 Virtual Interview Room Initialization

After the requirements have been determined, the system creates a virtual interview room equipped with Vapi AI voice technology and microphone access equipped with the WebRTC technology. The room serves as a synchronous interface of interaction between the human and the AI, which preconditions the possibility of having an immersive as well as real-time Q&A session between the two.

4.4 General Question Dataset Retrieval

The platform answers questions that are dynamically obtained by Hardy, that is created as a virtual store in Firebase, Fire store. This repository consists of HR, behavioral, aptitude and technical questions, which are filtered by the role and difficulty that are specified by the user. In contrast to the company-specific database, it is the base dataset from which a prompt during the live interview simulation is generated.

4.5 Interview Simulation Loop

Original work of the loan simulation loop is the functional center of VoxMentor. The AI voice agent will read every question in text form and the candidate will give a verbal response. Responses are recorded and transcribed by using OpenAI Whisper ASR. The system uses the extraction of linguistic, semantic, and prosodic features and utilizes LLM models as evaluators. Therefore, based on the data analysis, the flow controller automatically adjusts follow-up questions to make the dialogue comprehensive, interactive and adaptive that greatly resembles the dialogue of real interviews.

4.6 Automated Feedback and Recommendations

The transcripts and features extracted will be emailed at the end of each session to the feedback engine comprising of Google Gemini LLM and supervised scoring models. The candidate performance is assessed based on several criteria: technical knowledge, communication, problem-solving capacity and cultural fit, and confidence. Feedback is not only a quantitative score (0-100 scale) but, also qualitative assessment with recommendations on how to improve in a specific area.

4.7 Dashboard and History Management

Feedback saved in Firebase Firestore are presented in the user dashboard. The dashboard gives an overview of interview history, with a timeline view, summary of sessions and the scores of each attempt. Such a feature can make it possible to conduct longitudinal progress monitoring and allow data-based learning interventions.

4.8 Performance Reporting and Visualization

The system produces graphic report reports showing performance parameters in bar graphs, radar charts and trending graphs. Reports can be exported in PDF or CSV files, so that they can be reviewed offline or shared with mentors. These analytics assist the preparation process in the way of readiness to be tested systematically by candidates and trainers.

4.9 Aptitude and Coding Challenge Integration

Independent of the principal simulation session, VoxMentor provides an aptitude and coding test practice session. Aptitude template modules present inspection questions that can be graded automatically, whereas coding exercises run within an in-app Monaco editor with code execution running through the Judge0 API. Individual scores and explanations are saved and incorporated into the dashboard yet not part of the mock interview workflow.

4.8 Group Discussion Simulation Module (GD)

To evaluate collaborative and communication skills, VoxMentor incorporates a GD simulation environment. Multiple students, or a combination of students and AI-generated participants, are placed in a shared voice-enabled session. The AI moderates by introducing a topic (e.g., current affairs, technical scenario) and tracking participant contributions. Evaluation criteria include content relevance, clarity, teamwork, assertiveness, and leadership. Transcriptions are analyzed to provide both individual feedback and group dynamics assessment, enabling students to practice the skills required in recruitment-driven group discussions.

4.10 Company-Specific Preparation Repository

The other stand-alone module is the Company-Specific Repository, which offers specially curated material that concerns specific companies like Google, Amazon, or Microsoft. The repository contains earlier asked questions, HR requirements, role technical orientation as well as cultural engagement.

Unlike the overall database, it doesn't go into interview loop but it serves as an extra study aid for the candidates that go for specific companies.

V. RESULT & DISCUSSION

The created system, VoxMentor: A Voice-Driven AI System of Interview Simulation, Company-Specific Preparation, and Student Learning was tested with the help of the several types of functionalities such as interview generation, aptitude and coding tests, real-time feedback dashboards, and profile-based analytics. The results show that the platform has the potential to be an integrated learning and assessment environment where students and job seekers can take advantage of.

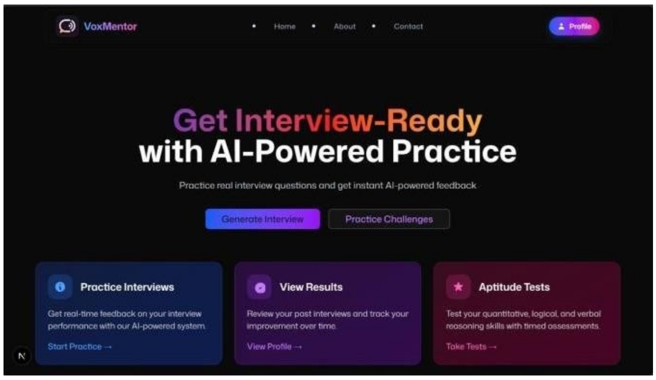

Fig.3 Home page

The set of experiments done on the proposed system VoxMentor include interview generation, aptitude tests, coding challenges and feedback dashboards. VAPI assistant (Fig. 1) was able to generate role- and skill-specific interview questions, and aptitude and coding modules (Figs. 2-3) allowed real-time assessments and showed an overall accuracy.

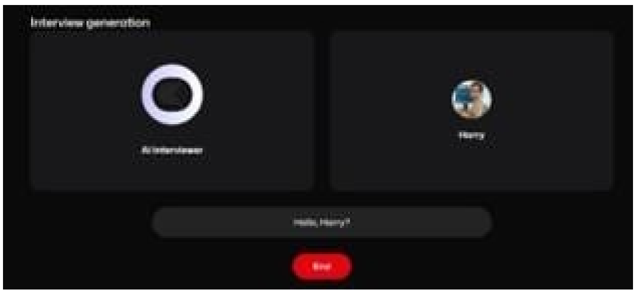

Fig.4 interview generation

Fig 4, This module dynamically generates interview questions based on user-defined parameters such as role, experience level, and technical domain using the integrated Google Gemini large language model.

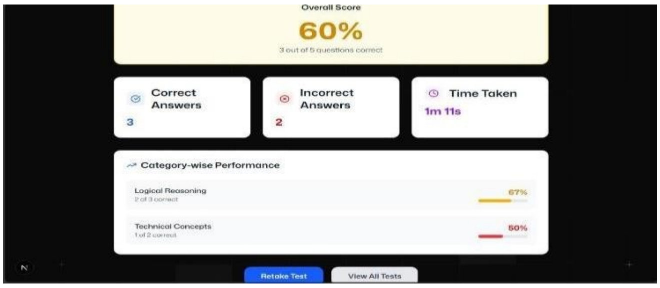

Fig.5 Aptitude & result

Fig 5, The aptitude evaluation interface displays assessment outcomes, highlighting accuracy, response time, and topic-wise performance to support analytical skill enhancement and readiness measurement.

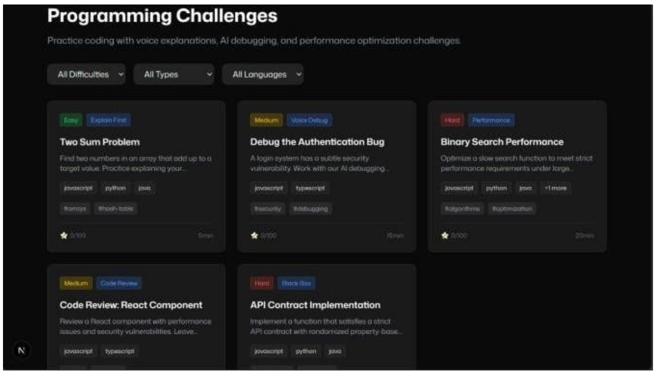

Fig.6 programming challenges

Fig 6, The coding challenge module enables users to attempt real-time programming tasks, which are automatically compiled and evaluated for correctness and efficiency using the Judge0 API framework.

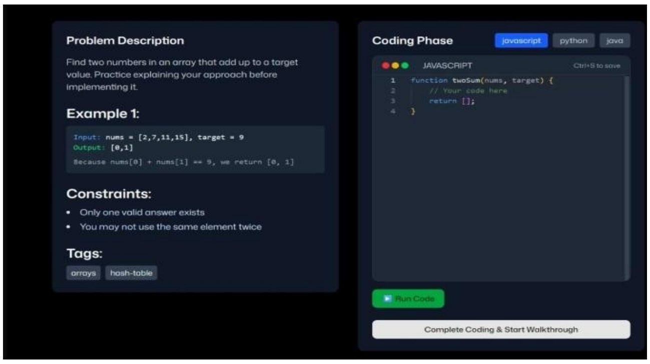

Fig.7 Monaco code editor

Fig 7, An integrated Monaco code editor facilitates writing, editing, and execution of code within the platform, supporting multi-language programming and real-time syntax validation.

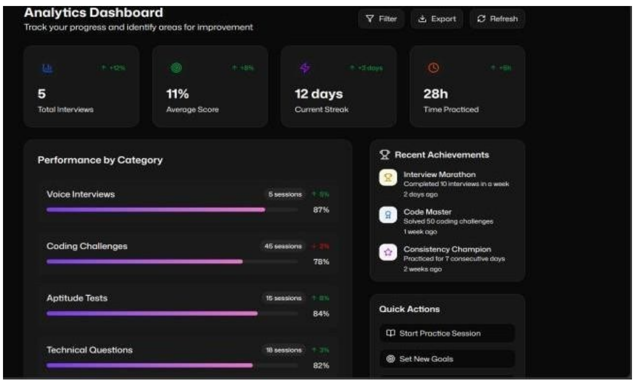

Fig.8 Student dashboard

Fig 8, The consolidated dashboard visualizes interview, aptitude, and coding results, enabling continuous tracking of student progress and identification of key improvement areas.

Fig.9 Interview feedback

This module dynamically generates interview questions based on user-defined parameters such as role, experience level, and technical domain using the integrated Google Gemini large language model (Fig.9).

Fig.10 Call Success Trend (VAPI)

Fig.10, illustrates the call success trend of the VAPI voice interaction module. The graph represents system performance during real-time interview sessions, showcasing the stability and reliability of AI-driven voice calls. The results demonstrate consistent connection success rates and low latency, confirming the robustness of the VAPI integration for uninterrupted candidate–AI interaction.

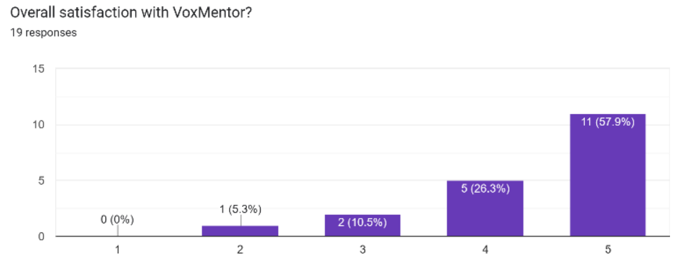

Fig.11 Overall User Satisfaction

Fig.11, Survey results showing overall user satisfaction with the VoxMentor platform. The majority of respondents (57.9%) rated their satisfaction at level 5, indicating high acceptance and positive user experience, while no respondents rated the system below level 2.

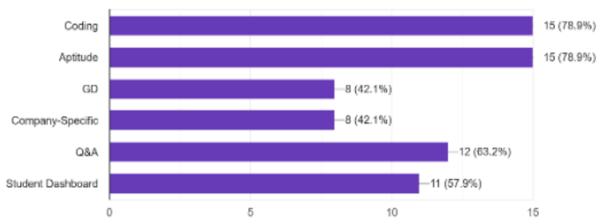

Fig.12 Skill Development Modules Chart

Fig.12, Skill development modules chart. The figure depicts participant responses identifying which VoxMentor modules were most useful for their learning experience. Coding (31.6%) and Aptitude (31.6%) modules were rated as the most beneficial, followed by Q&A simulation (26.3%), while Group Discussion (GD) and Company-Specific modules each received 21.1%.

VI. Conclusion and Future Scope

The proposed VoxMentor: A Voice-Driven Interview Simulation and Learning Platform demonstrates how artificial intelligence, speech recognition, and adaptive evaluation can work together to create a more realistic and engaging way to prepare for interviews. By combining voice-based mock interviews, instant feedback, and performance tracking, the system helps students strengthen both their communication and technical skills while building confidence for real-world recruitment scenarios. Vox Mentor's modular structure—which includes coding and aptitude challenges, group discussion (GD) practice, and company-specific preparation—offers learners a complete and flexible environment for skill development. Initial testing with students showed encouraging results, with improved engagement, better self-assessment, and greater satisfaction compared to traditional preparation methods. In the future, the platform will be enhanced with multilingual capabilities to support a wider range of learners, emotion and tone detection to evaluate behavioral aspects during interviews, and an expanded database of company-specific materials to stay aligned with industry trends. Advanced analytics and adaptive learning models will further personalize the experience, ensuring long-term growth and measurable improvement. With these advancements, VoxMentor has the potential to become a scalable, intelligent educational companion that supports students from classroom learning to successful employment.

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